

Venuste NDIKUMANA

FULL-STACK WEB & MOBILE DEVELOPMENT

Address: NR8, NM 4 Ave 16, Mpenge, Muhoza, Musanze, Rwanda
Phone: +250 799 375 874 **Email:** venustendikumana2003@gmail.com

GitHub: github.com/venusndk
Portfolio: portfolio-xlzv.onrender.com
LinkedIn: linkedin.com/in/ndikumana-venuste-985b4928a

PROFILE

Final-year Computer Science and Information Security undergraduate specializing in full-stack and mobile software engineering. Independently designs, builds, and ships complete production-style systems end-to-end from database schema and REST/real-time APIs to cross-platform Flutter clients while applying secure coding practices and integrating AI APIs into real projects.

Seeking the Ejo Labs Summer Talent Program 2026 to grow applied AI and product-engineering skills in a fast-moving, mission-driven team.

TECHNICAL SKILLS

Languages: Dart, JavaScript, TypeScript, Python, SQL

Mobile & Frontend: Flutter (BLoC architecture), React, Material 3, cross-platform UI

Backend & APIs: Node.js, Express.js, REST API design, Socket.IO, Django

Data & Cloud: PostgreSQL, SQLite, Firebase (Firestore, Admin SDK, Cloud Messaging)

AI Integration: OpenAI API (GPT-4o-mini) for real-time translation and applied NLP features

Security & Auth: JWT, bcrypt, rate limiting, input validation, Helmet.js, role-based access control

Practices & Tools: Git/GitHub, system design, Linux server deployment (Nginx, Gunicorn), Tailwind CSS

WORK EXPERIENCE

Full-Stack Developer Self-Directed Independent 2024 – Present

- Independently designed, built, and shipped multiple full-stack and mobile systems end-to-end — spanning backend APIs, real-time systems, relational and NoSQL database design, and cross-platform Flutter clients (see Selected Engineering Projects below).
- Applied secure coding practices (JWT authentication, bcrypt, input validation, rate limiting) and integrated third-party AI APIs (OpenAI) into production-style applications.

SELECTED FEATURED PROJECTS

[CampChat AI-Powered Multilingual Messaging Platform](#)

- Designed and built a secure, cross-platform real-time messaging app end-to-end: a Flutter/BLoC mobile client backed by a Node.js/Express API, with Socket.IO powering live 1-on-1 and group messaging, delivery/read receipts, typing indicators, and presence.
- Integrated the OpenAI API (GPT-4o-mini) to deliver on-demand and auto-translate messaging across English, French, Kinyarwanda, and Swahili solving a real cross-language communication gap for local users.
- Implemented a defense-in-depth security layer: JWT access/refresh token authentication, bcrypt password hashing, per-route rate limiting, parameterized SQL queries, Helmet.js headers, and express-validator input checks.
- Built an offline-first data layer using SQLite (sqflite) on-device synced against a normalized PostgreSQL schema, plus media sharing (image, video, audio, document) and a Material 3 light/dark UI.

[Driver Garage Linker \(DGL\) Driver Garage Marketplace Platform](#)

- Architected a multi-module cross-platform workspace connecting a Flutter mobile client (separate driver and garage-owner experiences) to a Node.js/Express backend, using Socket.IO for real-time trip/status updates and the Firebase Admin SDK for backend services.
- Modeled the core domain (drivers, garages, trips, vehicles) and designed the account-verification and role-based access flow connecting the two user types.
- Set up Firestore indexing and hosting configuration, and defined the project's lint/test quality-check workflow for both the Flutter client and the Node API.

[Personal Computer Maintenance System \(PCMS\) Full-Stack Service Management App](#)

- Built a full-stack web application, split into independent backend and frontend modules, to manage computer repair and maintenance service workflows covering customer intake, technician assignment, and service tracking.
- Combined a JavaScript service layer with Python-based components, applying separation-of-concerns between the presentation layer and business logic.

SORP Staff Operation & Reporting Platform (Final-Year Capstone, System Design)

- Authored a complete system design for an enterprise daily-duty and incident-reporting mobile platform: Firestore data schema, Flutter client architecture, Node.js API and Socket.IO event contracts, offline-first sync strategy, and a hierarchical role-based access control model. Repository setup pending.

EDUCATION

BSc. Computer Science and Information Security Class of 2027

University of Rwanda, College of Science and Technology · Focus: Full-Stack & Mobile Development

Secondary Education (A-Level) Combination: PCM (Physics, Chemistry, Mathematics)

Groupe Scolaire Sainte Bernadette de Save

Secondary Education (O-Level)

Groupe Scolaire Sainte Dominique Savio Kibangu

CERTIFICATIONS

Advanced Networking Operations Internet Society, 2026

AI Career Essentials ALX, September 2024

Junior Cybersecurity Cisco Networking Academy, August 2024

Introduction to Cybersecurity Cisco Networking Academy, May 2024

LANGUAGES

Kinyarwanda: Native · English: Intermediate · Swahili : Basic

REFERENCES

Available upon request.